

EE 252: Electrical Machines and Power Electronics

Lab (EMPEL)

Spring semester 2026

PE Converter making - Problem statements [20 Marks]

Solder the components given to realize the non-isolated DC-DC converters with the specifications provided as per the specifications provided below. (You are to use the TL 494 IC from for generating the pulses for the MOSFET. Solder the components on the TL494 board and test out the pulses (frequency and duty cycle variation) before providing the same to the converter. **Take suitable precautions such that the drive circuit ground and the power circuit ground are isolated!**)

You are required to simulate either a DC–DC converter (MATLAB) or an induction machine (ANSYS) for the given specifications. The simulation results must include all relevant waveforms and performance plots. In addition, a working hardware prototype of the converter/machine (as per the schedule provided) must be demonstrated during the final project evaluation.

Projects in Power Electronics

Topics for Batch 1

B1-G1: Buck Converter

- a) Specifications: Input 18 V, Output 12.5 V, Switching frequency 10 kHz, Output current 1 A.
Waveforms of Switch current and diode voltage in CCM.
- b) Reduce switching frequency to demonstrate DCM.

B1-G2: Boost Converter

- a) Specifications: Input 7.5 V, Output 12.5 V, Switching frequency 12 kHz, Output current 1 A.
Waveforms of Inductor current and switch voltage in CCM.
- b) Increase load resistance to demonstrate DCM.

B1-G3: Buck - Boost Converter

- a) Specifications: Input 15 V, Output 10-20 V, Switching frequency 7.5 kHz, Output current 1 A.
Waveforms of Switch current and inductor voltage in CCM.
- b) Reduce switching frequency to demonstrate DCM.

B1-G4: Boost Converter

- a) Specifications: Input 10 V, Output 20 V, Switching frequency 5 kHz, Output current 1 A.

Waveforms of diode current and inductor voltage in CCM.

b) Increase load resistance to demonstrate DCM.

B1-G5: Buck Converter

a) Specifications: Input 10 V, Output 2-8 V, Switching frequency 20 kHz, Output current

Waveforms of inductor current and switch voltage in CCM.

b) Reduce switching frequency to demonstrate DCM.

B1-G6: SEPIC Converter

a) Specifications: Input 15 V, Output 10-20 V, Switching frequency 5 kHz, Output current 1 A.

Waveforms of switch current and diode voltage in CCM.

b) Increase load resistance to demonstrate DCM.

B1-G7: Buck - Boost Converter

a) Specifications: Input 15 V, Output 12-18 V, Switching frequency 12.5 kHz, Output current 1 A.

Waveforms of diode current and inductor voltage in CCM.

b) Reduce switching frequency to demonstrate DCM.

B1-G8: Buck Converter

a) Specifications: Input 10 V, Output 5 V, Switching frequency 15 kHz, Output current 1 A.

Waveforms of switch current and diode voltage in CCM.

b) Increase load resistance to demonstrate DCM.

B1-G9: Buck - Boost Converter

a) Specifications: Input 20 V, Output 15-25 V, Switching frequency 10 kHz, Output current 1 A.

Waveforms of diode current and switch voltage in CCM.

b) Reduce switching frequency to demonstrate DCM.

B1-G10: Cuk Converter

a) Specifications: Input 15 V, Output 10-20 V, Switching frequency 7.5 kHz, Output current 1 A.

Waveforms of inductor current and diode voltage in CCM.

b) Increase load resistance to demonstrate DCM.

Topics for Batch 2

B2-G1: Buck Converter

- a) Specifications: Input 12.5 V, Output 7.5V, Switching frequency 12.5 kHz, Output current 1 A.
Waveforms of Switch current and diode voltage in CCM.
- b) Reduce switching frequency to demonstrate DCM.

B2-G2: Boost Converter

- a) Specifications: Input 14 V, Output 20 V, Switching frequency 7.5 kHz, Output current 1 A.
Waveforms of Inductor current and switch voltage in CCM.
- b) Increase load resistance to demonstrate DCM.

B2-G3: Buck - Boost Converter

- a) Specifications: Input 12.5 V, Output 8-15 V, Switching frequency 20 kHz, Output current 1 A.
Waveforms of Switch current and diode voltage in CCM.
- b) Reduce switching frequency to demonstrate DCM.

B2-G4: Cuk Converter

- a) Specifications: Input 10 V, Output 7.5-12.5 V, Switching frequency 10 kHz, Output current 1 A.
Waveforms of diode current and inductor voltage in CCM.
- b) Increase load resistance to demonstrate DCM.

B2-G5: Buck Converter

- a) Specifications: Input 20 V, Output 7.5 V, Switching frequency 20 kHz, Output current 1 A.
Waveforms of inductor current and switch voltage in CCM.
- b) Reduce switching frequency to demonstrate DCM.

B2-G6: Buck - Boost Converter

- a) Specifications: Input 15 V, Output 8-18 V, Switching frequency 12.5 kHz, Output current 1 A.
Waveforms of switch current and diode voltage in CCM.
- b) Increase load resistance to demonstrate DCM.

B2-G7: Zeta Converter

- a) Specifications: Input 12 V, Output 6-18 V, Switching frequency 10 kHz, Output current 1 A.
Waveforms of diode current and inductor voltage in CCM.
- b) Reduce switching frequency to demonstrate DCM.

B2-G8: Boost Converter

- a) Specifications: Input 7.5 V, Output 12.5 V, Switching frequency 10 kHz, Output current 1 A.
Waveforms of switch current and diode voltage in CCM.
- b) Increase load resistance to demonstrate DCM.

B2-G9: Zeta Converter

- a) Specifications: Input 10 V, Output 8-15 V, Switching frequency 7.5 kHz, Output current 1 A.
Waveforms of diode current and switch voltage in CCM.
- b) Reduce switching frequency to demonstrate DCM.

B2-G10: Buck Converter

- a) Specifications: Input 25 V, Output 17.5 V, Switching frequency 15 kHz, Output current 1 A.
Waveforms of inductor current and diode voltage in CCM.
- b) Increase load resistance to demonstrate DCM.

**Final Exam of
Hardware prototype/ making of a DC-DC Converter + Report: 1 / batch
20 Marks**

**Apr. 7th 2026 for B1 – all groups
&
Apr. 10th 2026 for B2 – all groups !**

Projects in Electrical Machines (For interested students only)

Hardware : Design the single layer winding for an Induction Motor (IM) with respect to different pole combinations with open-end windings.

The winding design problem statement for different batches is,

Q1 : 115 V_{rms} per phase, 24 Slot, 250 W, 3-phase, 6-pole, fractional slot full pitch winding

Q2 : 115 V_{rms} per phase, 24 Slot, 200 W, 3-phase, 2-pole, full pitch winding

Q3 : 115 V_{rms} per phase, 24 Slot, 200 W, 3-phase, 4-pole, full pitch winding

Q4 : 115 V_{rms} per phase, 24 Slot, 250 W, 3-phase, 2-pole, short pitch by 1 slot

Q4 : 115 V_{rms} per phase, 24 Slot, 200 W, 3-phase, 4-pole, short pitch by 1 slot

Tasks for Hardware Implementation are,

1. Disassemble the given machine, and note down the design data for a given machine, like, number of slots, stator inner diameter, outer diameter, shaft diameter, air gap, length of the core, slot area, slot depth, stator back iron length etc.,
2. Draw the winding diagram as per the given problem statement.
3. Calculate the number of turns, number of coils/phase, current rating, SWG, wire length, winding factor, copper weight, slot fill factor, winding resistance, etc.,
4. Copper savings with the short pitch winding over a full pitch winding.
5. Make the coils using the winding-making machine.
6. Insert the coils into the slots one by one.
7. Complete the winding of a machine.
8. Do the short circuit test with the machine body and the continuity test of each phase.
9. Assemble the machine and complete the testing of the machine with a 3-phase auto Transformer.
10. Verify the speed with Tachometer, phase voltages and phase currents waveforms.
11. Check the phase resistance and compare it with the theoretical resistance.

ANSYS Simulation : Design and magnetic transient analysis of IM using ANSYS electromagnetic software.

Tasks for simulation implementation are,

12. Implement the same winding in ANSYS Rmxprt.
13. Convert the ANSYS Rmxprt project to ANSYS Maxwell environment and analyse the transient analysis.
14. Perform the transient analysis for torque input, speed input and efficiency maps.
15. Compare practical machine design data with the theoretical design. Give your comments.

Reference:

- 1) Thomas A. Lipo, Introduction to AC Machine Design, IEEE Press, John Wiley & Sons, Inc., NJ, USA.
- 2) Juha Pyrhonen, Tapani Jokinen, Valeria Hrabovcova, Design of Rotating Electrical Machines, Second Edition, John Wiley & Sons Ltd, West Sussex, UK.
- 3) https://www.youtube.com/watch?v=zisdVkYwnEI&list=PLgMDNELGJ1CaohjewgMtnZb_ngBSZk0uv&index=29 (*Lec 28 Lab Session on Re-winding of Induction Motor (Example: Double Layer Winding)*)

Notes on the Driver board / components

1. Drive Circuit: The TL494 PWM generator IC

We will be using the TL494 board, for generating the pulses for the DC-DC PE converters. The datasheets for all the components have been attached in the same drive link. Do read the datasheets of TL494 and the MOSFET, DIODE and the driver IC TC4428A thoroughly. The circuit diagram and the PCB Schematics have also been provided in the attachments. The first step is to solder the components carefully, **solder the IC holder first and not the TL494 directly !**

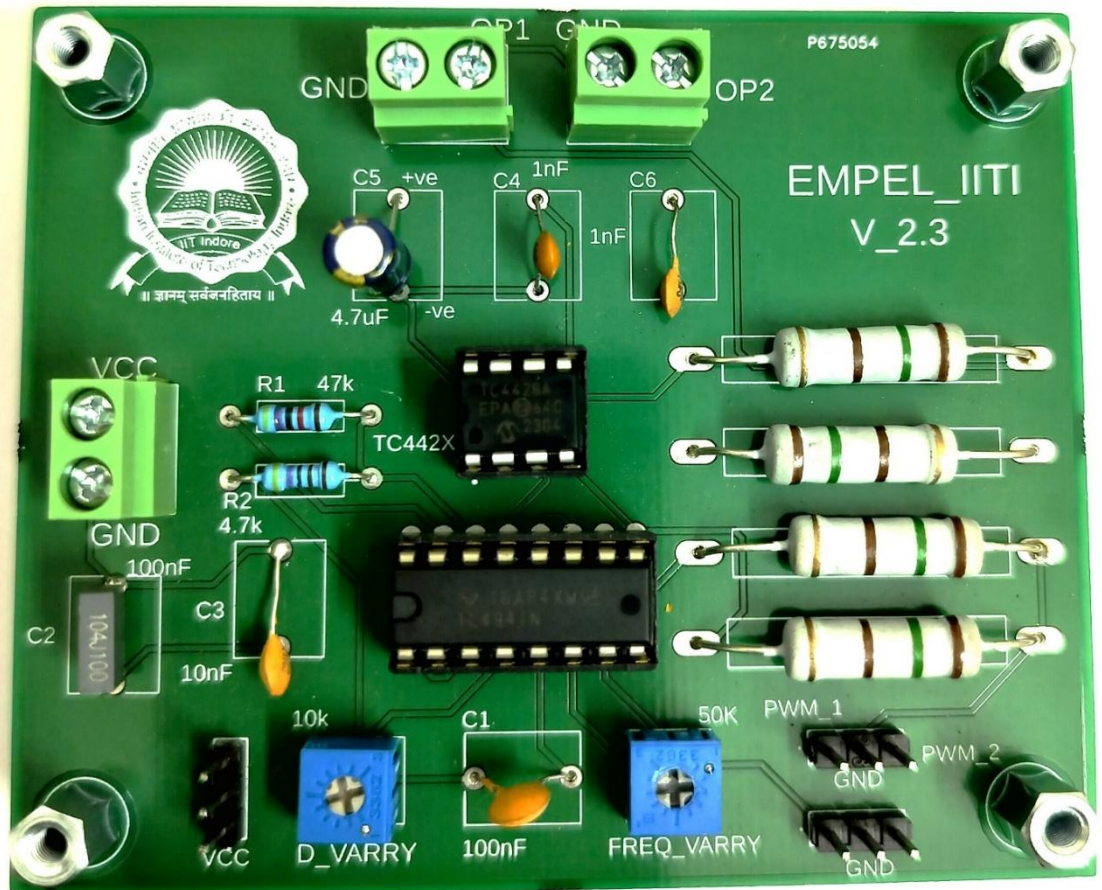


Fig: Schematic and Sample pic of board after soldering the components

The board has to be used in the emitter follower mode. Using the jumper wires connect the pins appropriately and check the pulses at the output pins by connecting them to an oscilloscope using voltage probes. The potentiometers D-vary and F-vary can be adjusted to change the duty cycle and switching frequency respectively. The same can be set to the values mentioned in the problem statements.

After checking the gating pulses at the output of the driver IC, you can proceed to provide the pulses to the power MOSFET gate and source terminals. Ensure that the rest of the power circuit is soldered and connected up properly as per the desired converter configuration before this is done. ***You may test the circuit on the Perf/ Universal board provided, however, the final demo should be with a PCB fabricated in the Makerspace Lab.***

The details have been provided:

Qty	Value	Device	Package	Parts	Description
2	10 k Ω , R3 50 k Ω	3362P potentiometers	3362P	10k (D-Vary), 50k (F-Vary)	Potentiometer 3362 Series
3		Green connector terminals			Connector terminals for input power supply and mosfet G-S
9		PINHD-1X2	1X02		Berg strip pins header
1	100 nF	C10/4	C10B4	C1	Tantalum Capacitor
1	100 nF			C2	Box Film Capacitor
1	10 nF	C2.5/2	C2.5-2	C3	Tantalum Capacitor
2	1 nF	C5/7.2	C5B7.2	C4, C6	Tantalum Capacitor
1	4.7 μ F, 63 V			C5	Electrolytic Capacito
2	150 Ω 2 W, CFR	R- EU_0617/22	0617/22	R3-R6	RESISTORS
1	4.7 k Ω $\frac{1}{4}$ W MFR	R- EU_0207/10	0207/10	R2	RESISTOR
1	47 k Ω $\frac{1}{4}$ W MFR	R- EU_0207/10	0207/10	R1	RESISTOR
1	TL494IN	TL494IN	DIL16	IC1	SWITCHMODE Pulse Width Modulation Control Circuit
1	TC4428A	TC4428A	DIL8	IC2	Mosfet driver IC

2. Power Circuit: The other components

Power Electronic switches:

The power **MOSFET** provided is IRFZ44NPbF from International Rectifiers (IR) having a maximum current rating of 49 A and can block 55 V. It has to be mounted on the **heat sink** after applying the heat sink paste (and then screwed to the heat sink).

The **Power Diode** provided is QH08TZ600 a hyperfast switching diode from Power Integrations having a maximum current rating of 8 A and can block 600 V. This can be soldered directly.

The power supply for the complete gate drive – TL494 board and the MOSFET driver board may be set at +12 V. The ground of the DC-DC converter is to be different - ISOLATED.

Filter components:

- The inductance given is a 1 mH toroidal inductor with a maximum current rating of 2.4 A.
- Ensure that while testing the load rheostat resistance is set such that the current through

the inductor never exceeds 1.5 A!

- The electrolytic capacitances given are a 470 μF , 63 V Capacitor (for the output) and A 100 μF , 25 V . Be careful about the polarity while connecting it up (especially with Buck-Boost configuration!)

The load resistance is a variable rheostat of 40 Ω and 3 A capacity.

3. The Power circuits and Isolation

You must take due caution in the aspect of isolation of the gate drive supply and the power circuit grounds.

1) Since the oscilloscopes in the lab have **no internal electrical isolation**, and there are no differential voltage probes available in the lab. You should only see one voltage at a time in the scope. If simultaneously two or more voltages have to be seen, use differential probes !

2) Depending on the problem specifications, you have to display currents through the various elements. For this you can use the 0.01 Ω , 3 W metal resistors in series with the corresponding elements in the board. The terminals need to be filed for making the connections. There are 2 current probes in the lab, so you need to share the same.

3) For easy implementation, the basic DC-DC Converters may be connected up as shown in the following figures; the components need to be soldered up on the general-purpose PCB:

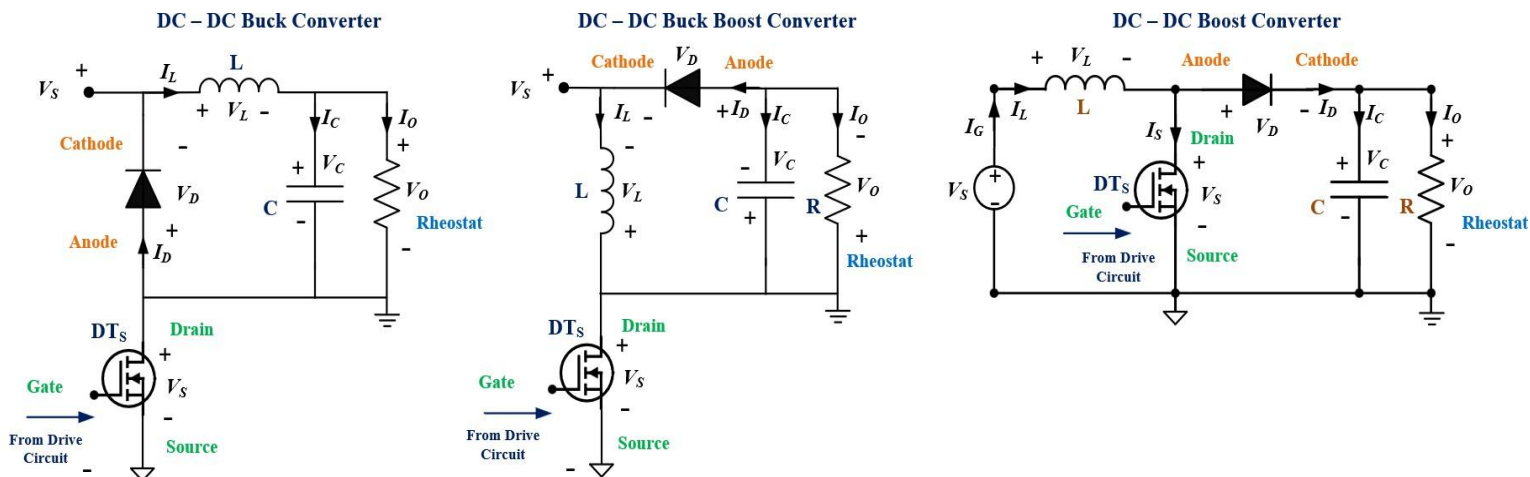


Fig: Power Circuits for the DC-DC Converters
(On similar lines for the Cuk, SEPIC and Zeta converters)

4) Check your circuit operation with load connected. Set the rheostat value to the proper resistance and then connect the same at the load side. Do not switch on the power without the load connected !

You are to record all the waveforms and submit a **1 page typed report** (PDF) containing the following:

- Mention Set and Batch no., Roll no of students and problem statement.
- Picture of the soldered gate driver and converter boards
- Snapshots of the waveforms in CCM and DCM.

Clearly show the measurements of time, pk-pk voltages/ currents and the frequency in the waveform snapshots using “measure” feature.

All the best for the projects !!

*EMPEL thankfully acknowledges the members of the Electronics Club for
their support in designing and testing of the driver PCB*
